

EECS 836: Machine Learning

Zijun Yao

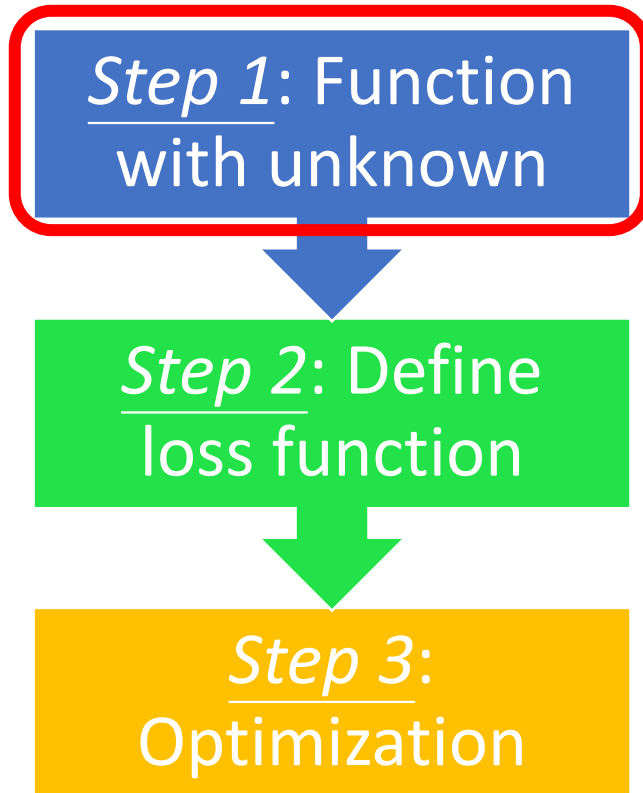
Assistant Professor, EECS Department
The University of Kansas

Agenda

- Introduction to meta-Learning

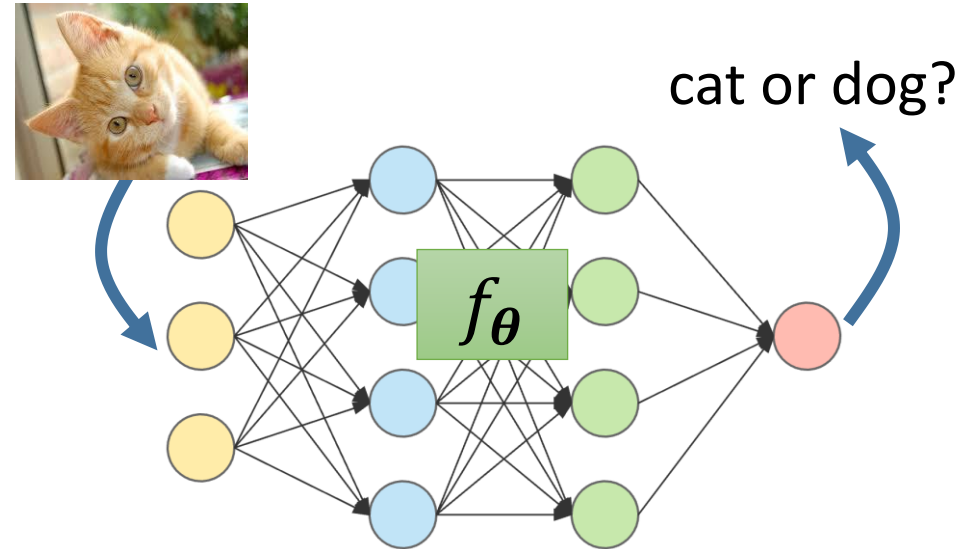
Machine Learning

= Looking for a function



Dog-Cat Classification

$$f(\text{image of cat}) = \text{"cat"}$$



Weights and biases of neurons are unknown parameters (*learnable*).

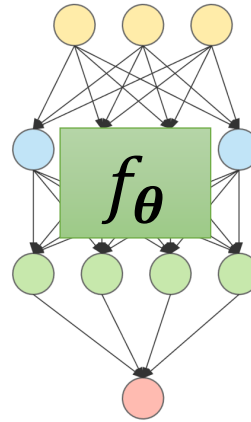
Using θ to represent the unknown parameters.

Machine Learning

Step 1: Function with unknown

Step 2: Define loss function

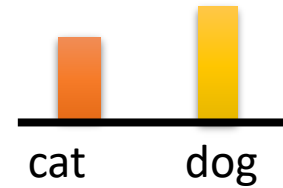
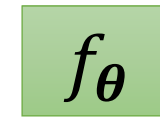
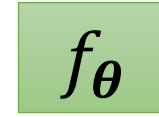
Step 3: Optimization



$$L(\theta)$$

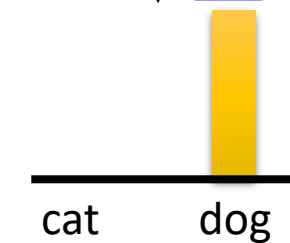
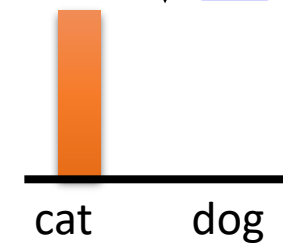
$$L(\theta) = \sum_{k=1}^K e_k$$

Training Examples



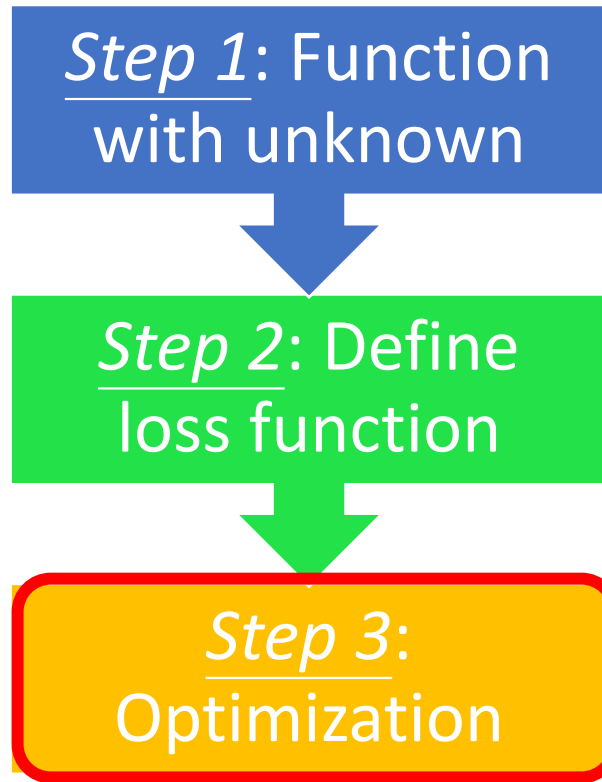
Cross-entropy e_1

e_2



Ground Truth

Machine Learning 101



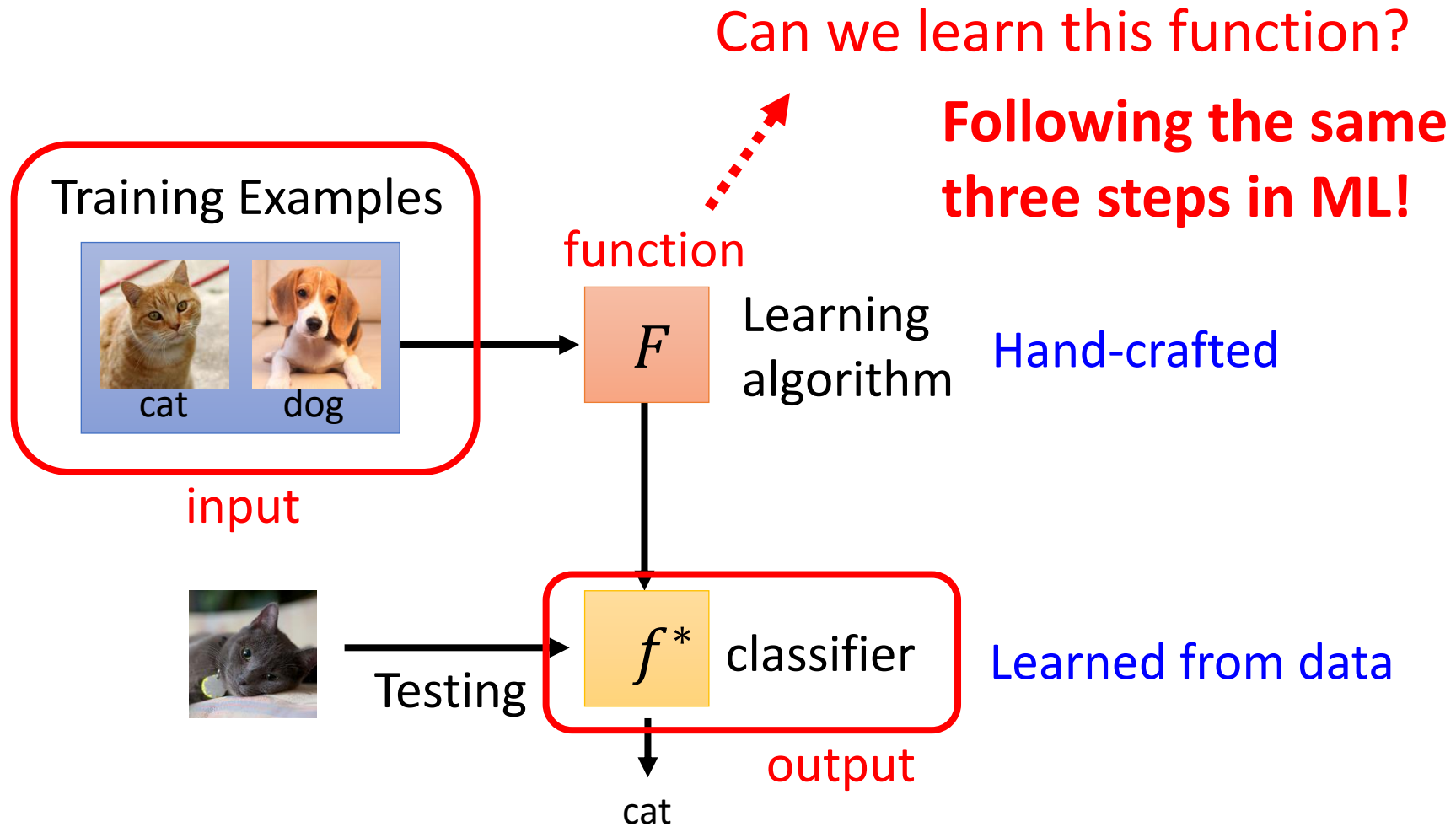
loss: $L(\boldsymbol{\theta}) = \sum_{k=1}^K e_k$ sum over training examples

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$$

done by gradient descent

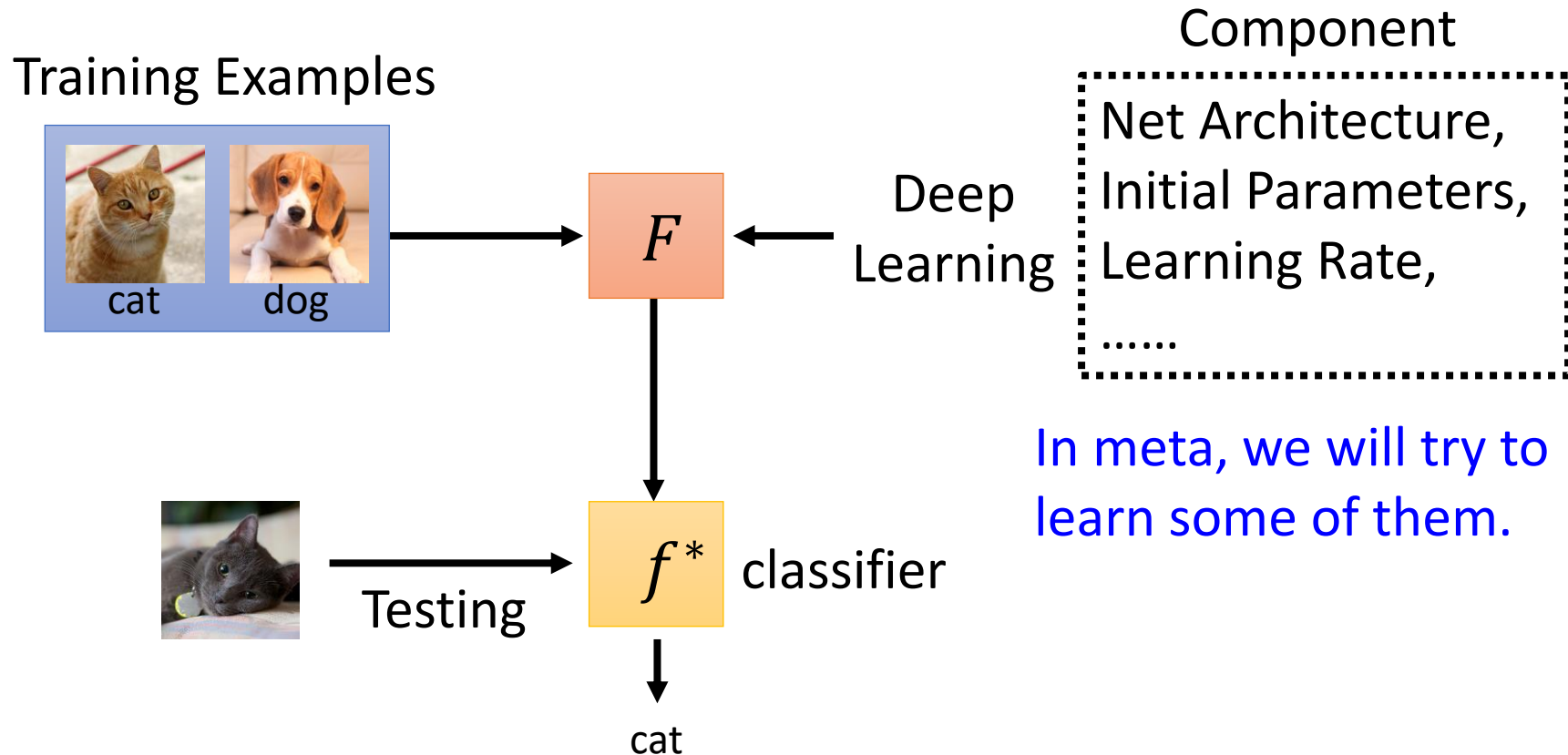
$f_{\boldsymbol{\theta}^*}$ is the function learned by learning algorithm from data

What is Meta Learning?



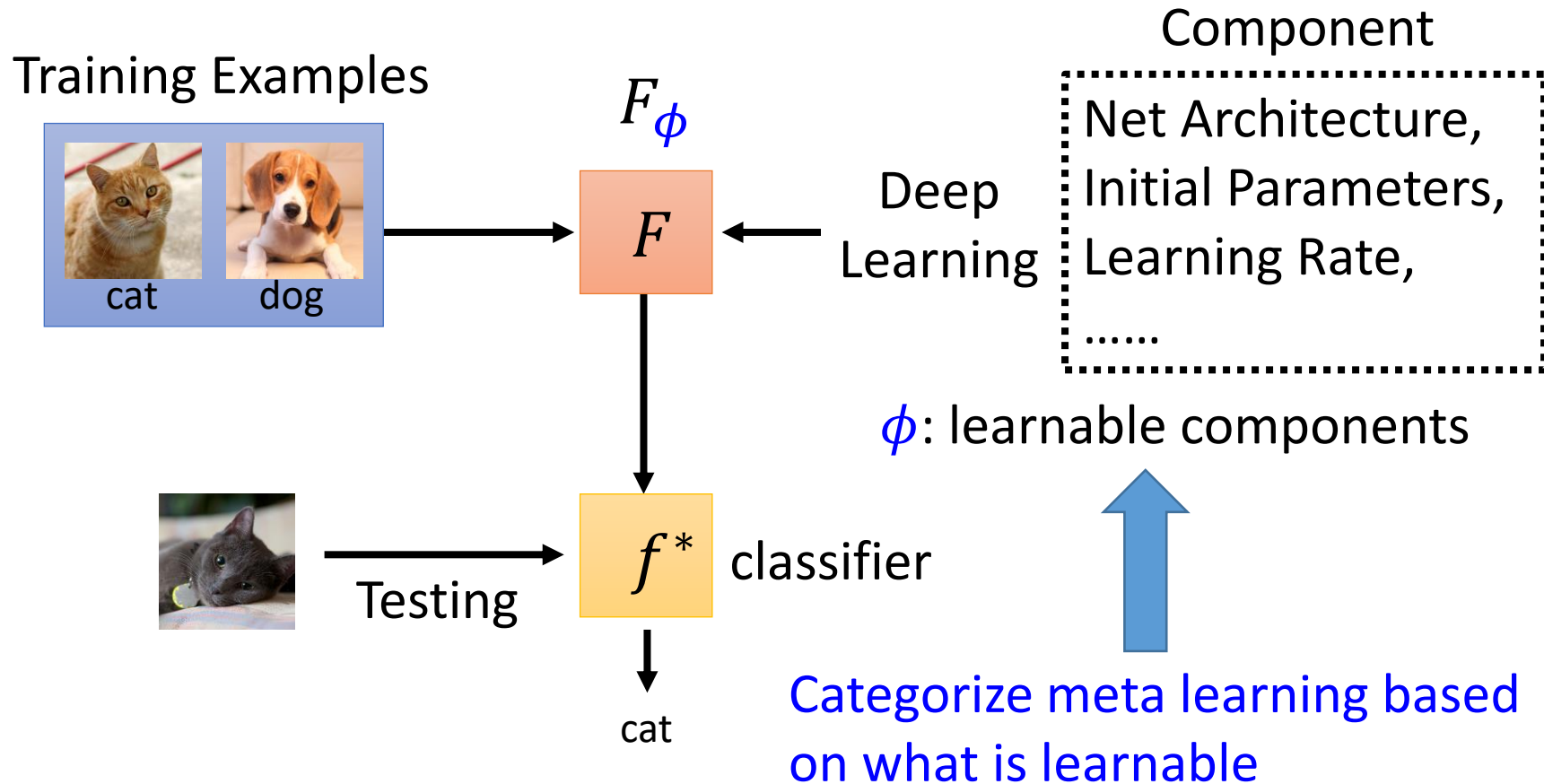
Meta Learning – Step 1

- What is **learnable** in a learning algorithm?



Meta Learning – Step 1

- What is **learnable** in a learning algorithm?



Meta Learning – Step 2

- Define loss function for learning algorithm F_ϕ

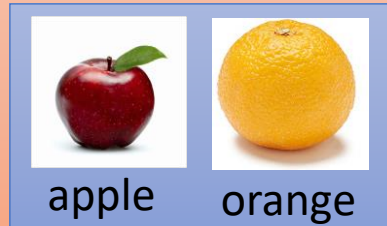
$$L(\phi)$$



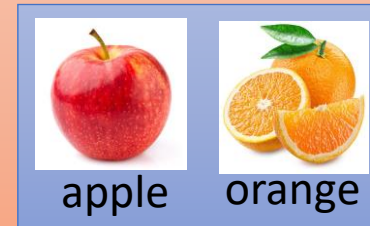
Training Tasks

Task 1
Apple &
Orange

Train

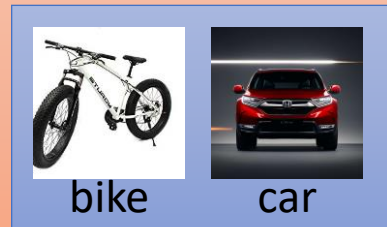


Test

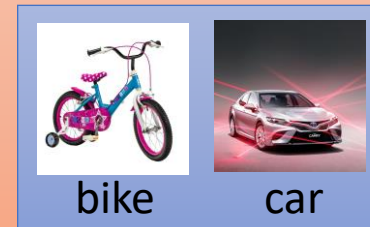


Task 2
Car & Bike

Train



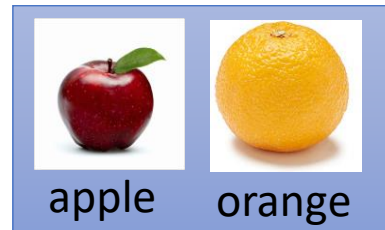
Test



Meta Learning – Step 2

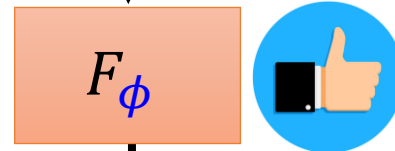
Task 1

Training
Examples

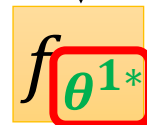


How to define $L(\phi)$

$L(\phi)$ ↓



classifier

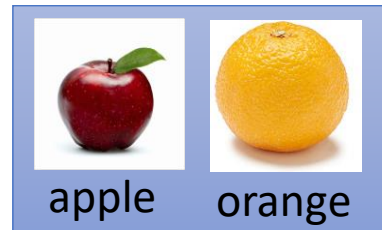


θ^{1*} : parameters of the classifier learned by F_ϕ
using the training examples of task 1

Meta Learning – Step 2

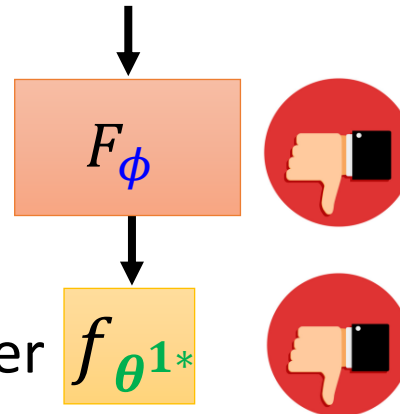
Task 1

Training
Examples



How to define $L(\phi)$

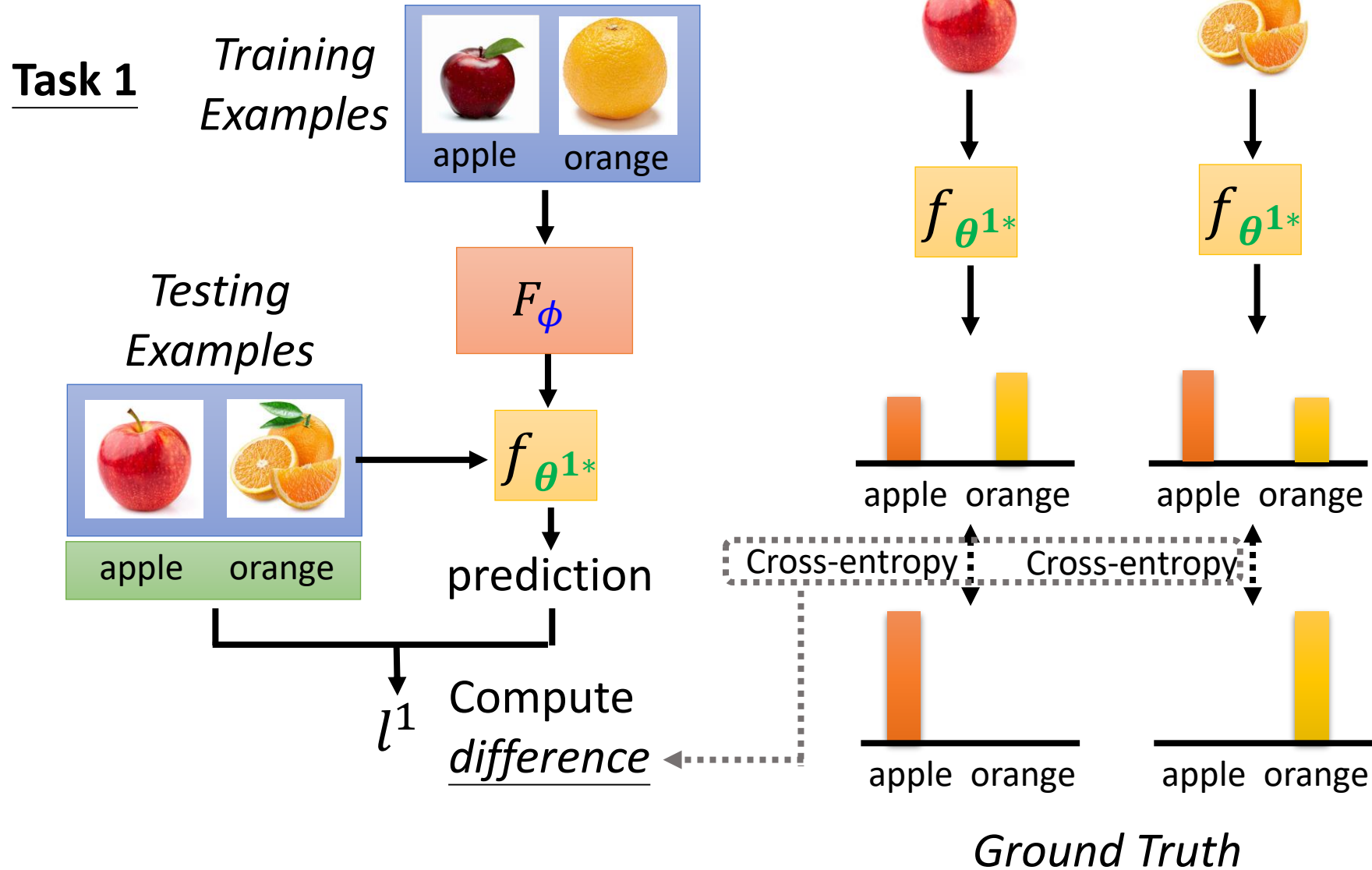
$$L(\phi) \uparrow$$



How can we know a classifier is good or bad?

Evaluate the classifier on testing set

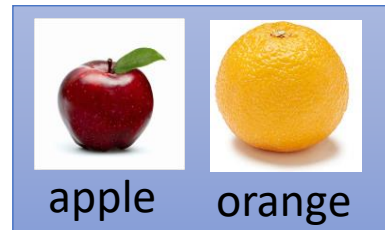
Meta Learning – Step 2



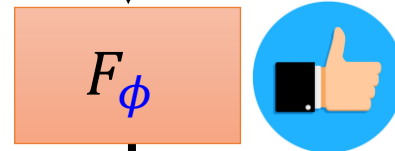
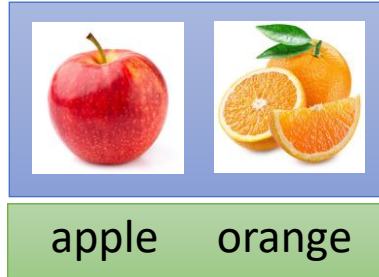
Meta Learning – Step 2

Task 1

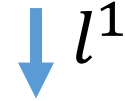
Training
Examples



Testing
Examples

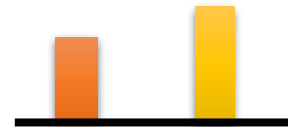
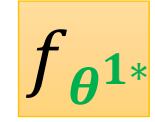
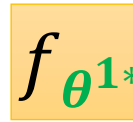


prediction



Compute
difference

Testing Examples



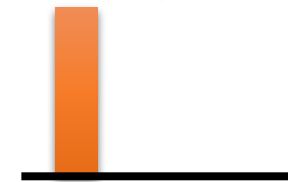
apple orange



apple orange

Cross-entropy

Cross-entropy



apple orange



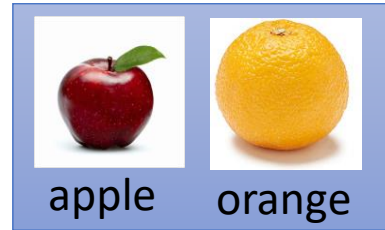
apple orange

Ground Truth

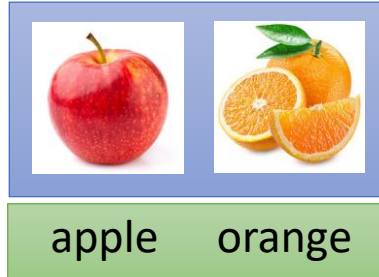
Meta Learning – Step 2

Task 1

Training
Examples



Testing
Examples



F_ϕ



$f_{\theta^{1*}}$



prediction



l^1

Compute
difference

Testing Examples



$f_{\theta^{1*}}$



apple orange



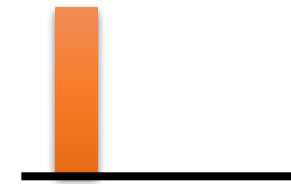
$f_{\theta^{1*}}$



apple orange

Cross-entropy

Cross-entropy



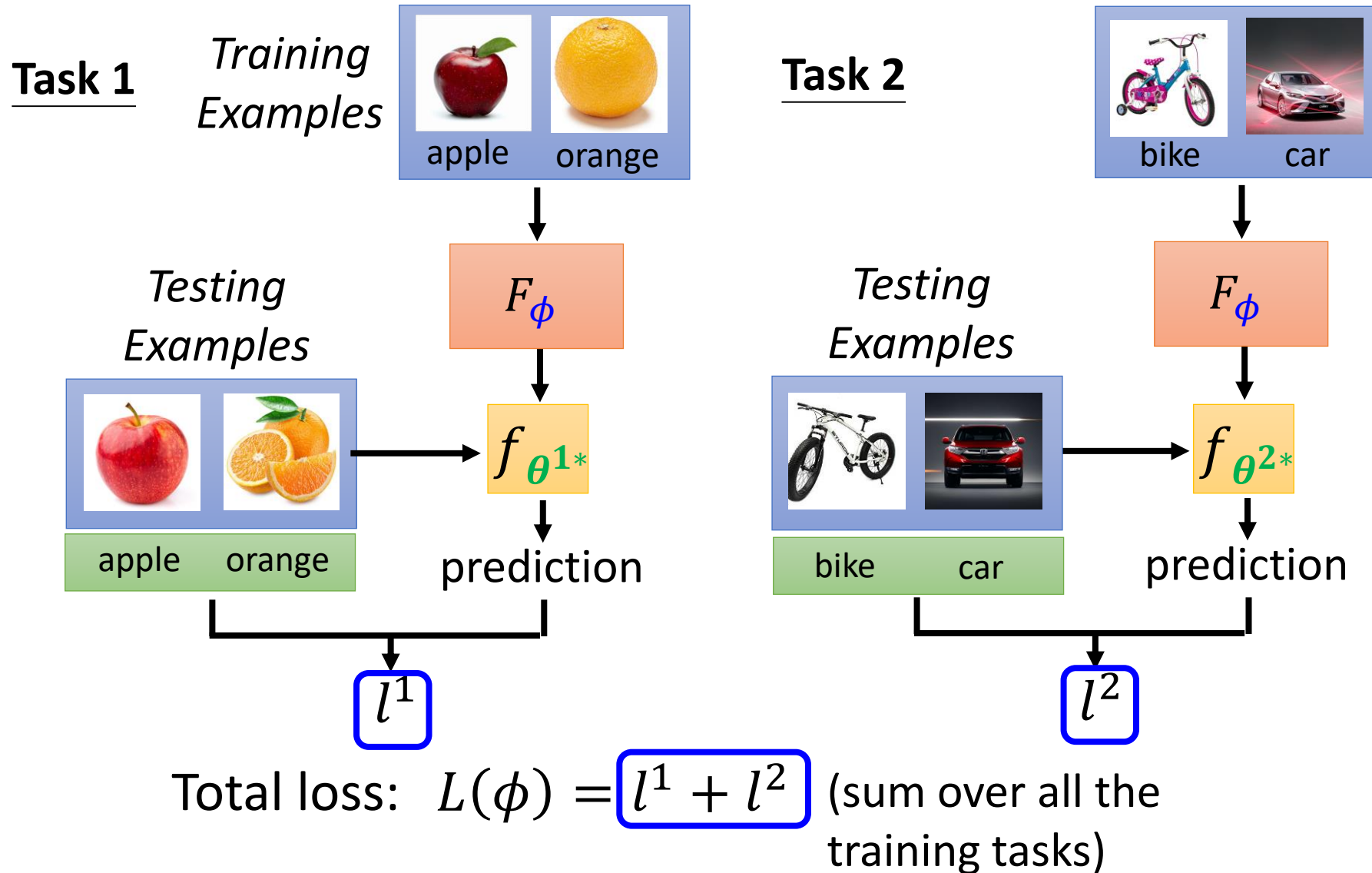
apple orange



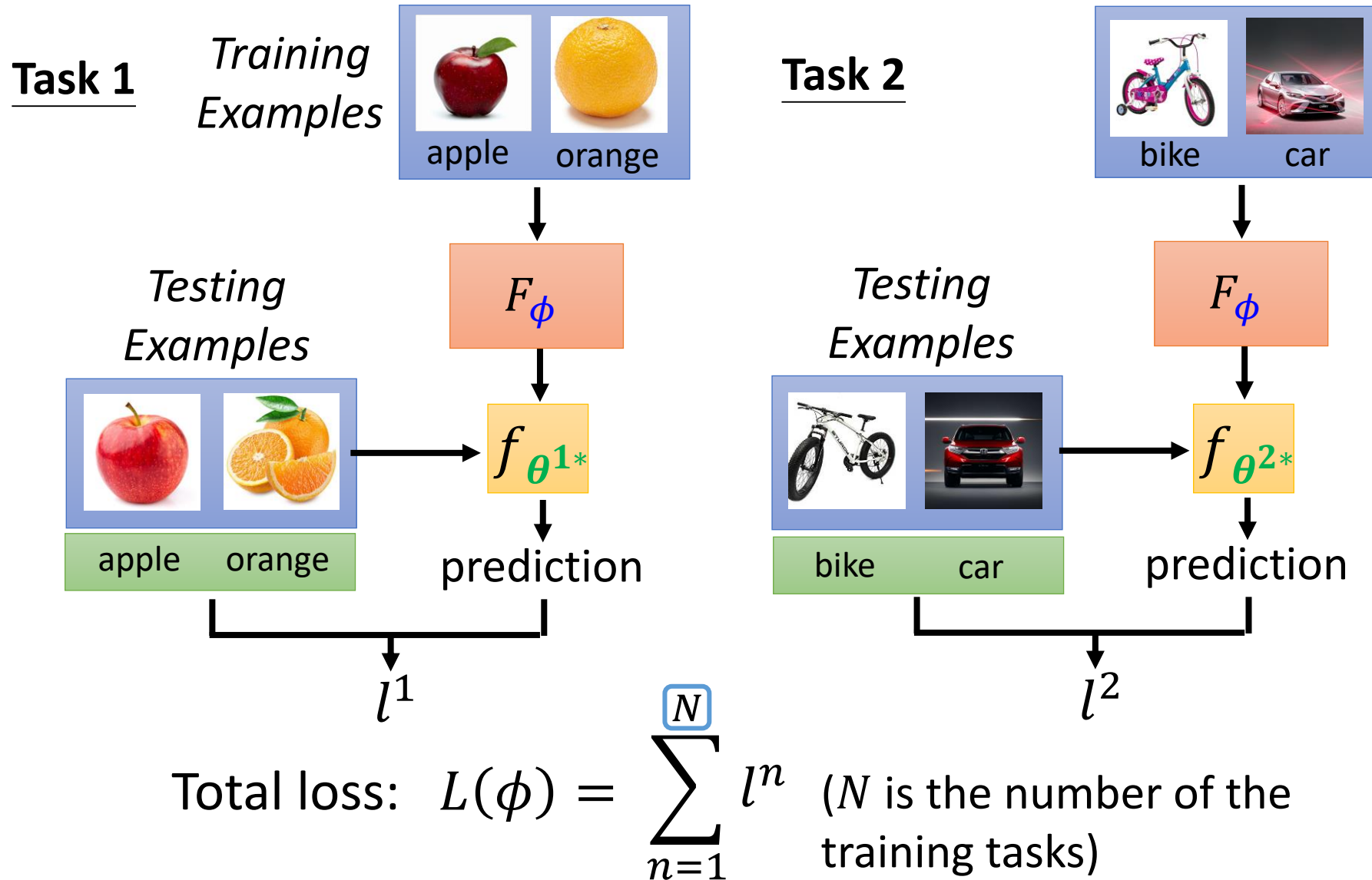
apple orange

Ground Truth

Meta Learning – Step 2



Meta Learning – Step 2



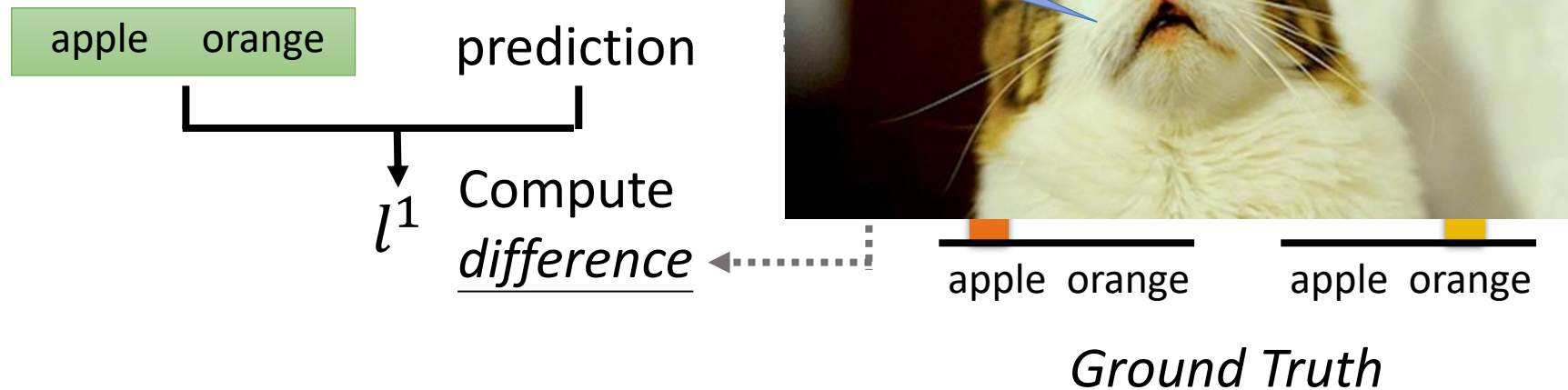
Meta Learning – Step 2

Task 1

In typical ML, you compute the loss based on **training examples**

In meta, you compute the loss based on **testing examples**

Hold on! You use **testing examples** during training???



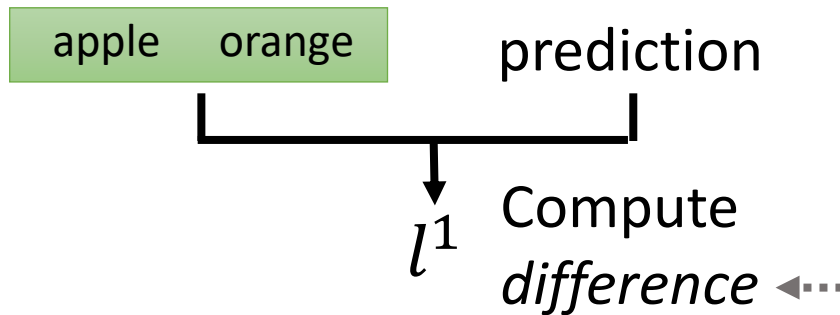
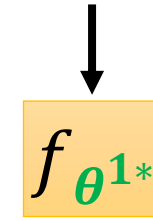
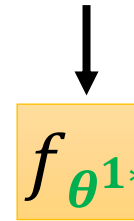
Meta Learning – Step 2

Task 1

In typical ML, you compute the loss based on **training examples**

In meta, you compute the loss based on **testing examples** of **training tasks**.

Testing Examples



Meta Learning – Step 3

- Loss function for learning algorithm

$$L(\phi) = \sum_{n=1}^N l^n$$

- Find ϕ that can minimize $L(\phi)$

$$\phi^* = \arg \min_{\phi} L(\phi)$$

- Using the optimization approach you know

If you know how to compute $\partial L(\phi) / \partial \phi$

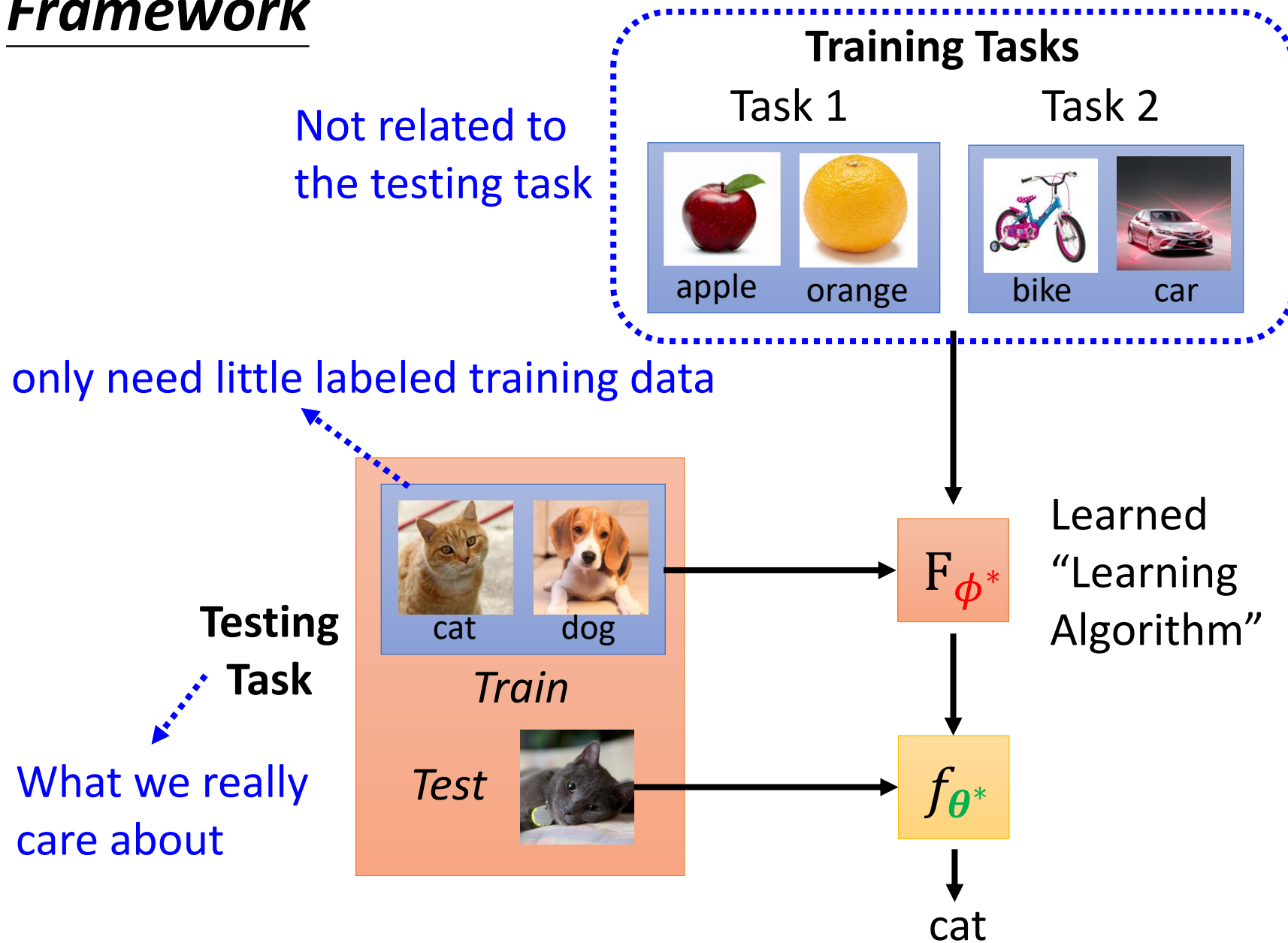
Gradient descent is your friend.

What if $L(\phi)$ is not differentiable?

Reinforcement Learning / Evolutionary Algorithm

Now we have a learned “learning algorithm” F_{ϕ^*}

Framework



ML v.s. Meta

Goal

Machine Learning $\approx$ find a function f

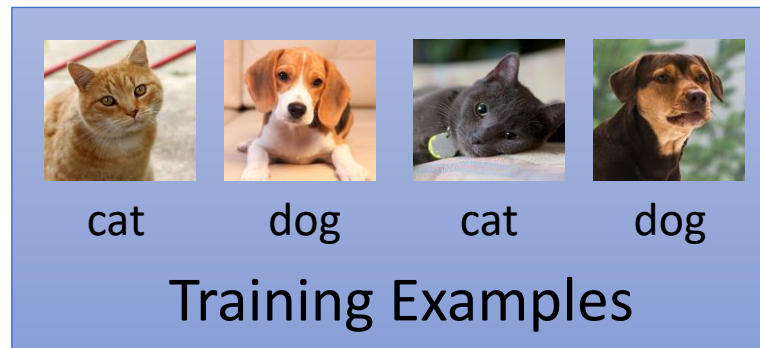
Dog-Cat
Classification

$$f(\text{img}) = \text{"cat"}$$

Meta Learning

$\approx$ find a function F that finds a function f

Learning
Algorithm F



$$) = f$$

Training Data

Machine Learning

One task



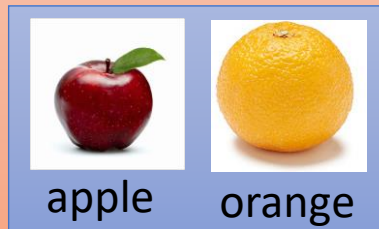
Train

Meta Learning

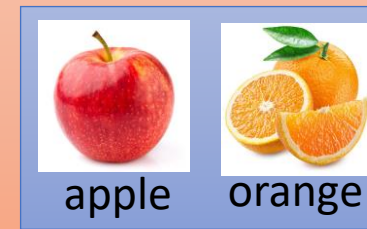
Training tasks

Task 1
Apple &
Orange

Train

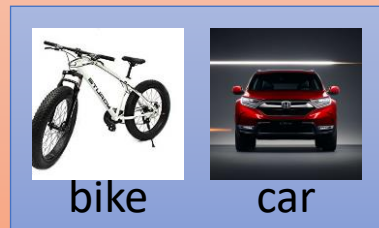


Test

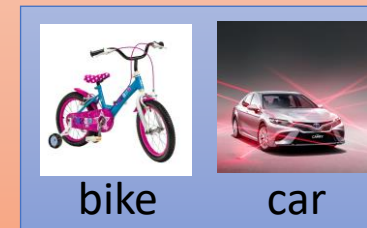


Task 2
Car & Bike

Train



Test



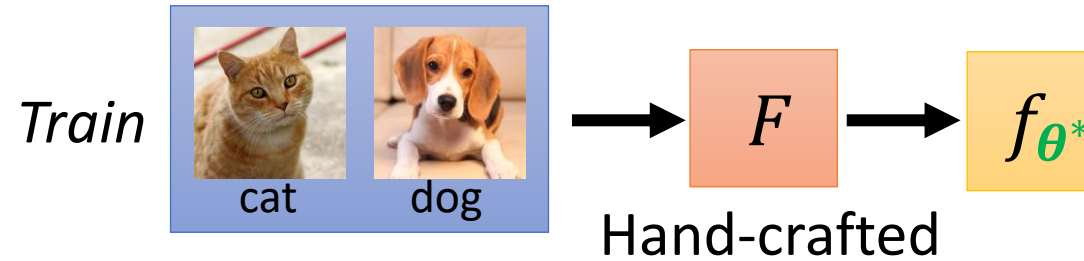
Support set

Query set

(in the literature of “*learning to compare*”)

Machine Learning

Within-task Training



Meta Learning

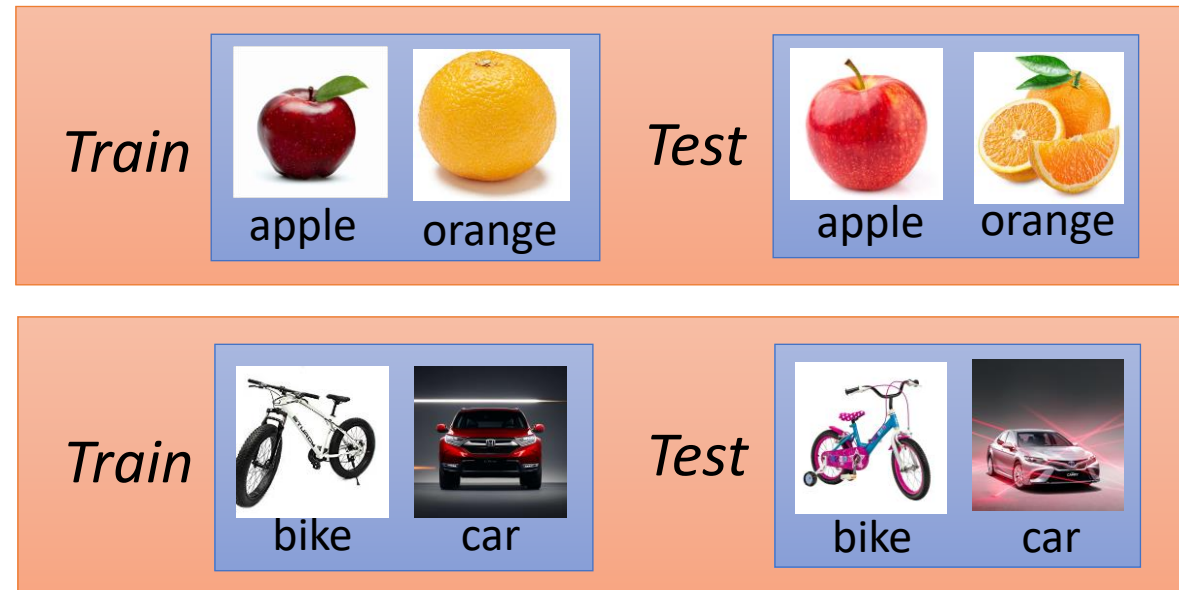
Training Tasks



F_{ϕ^*} Learning Algorithm

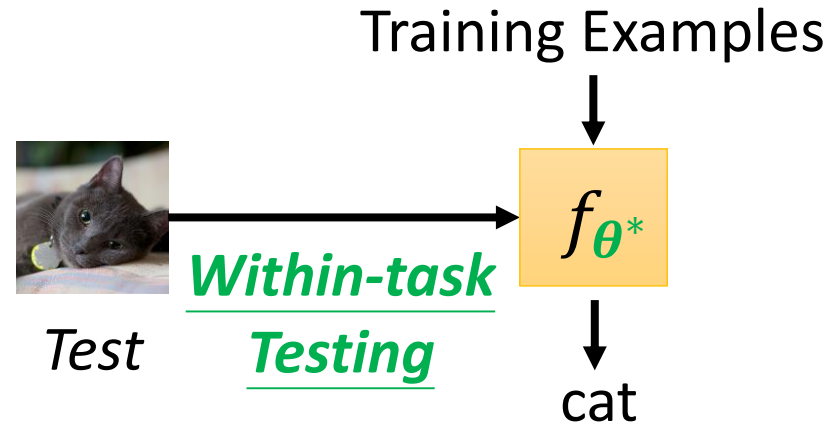
Task 1

Task 2

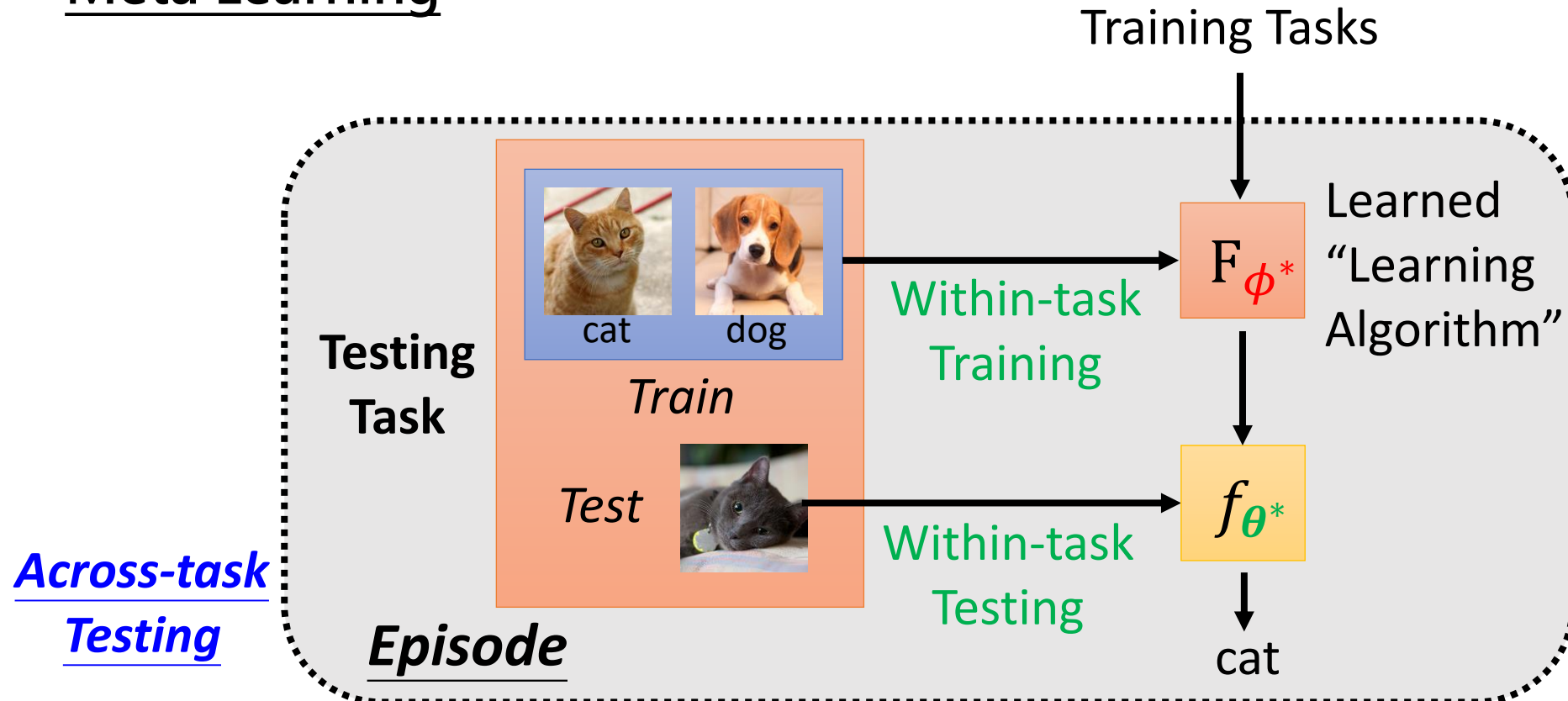


Across-task Training

Machine Learning



Meta Learning



Loss

Machine Learning

$$L(\boldsymbol{\theta}) = \sum_{k=1}^K e_k$$

Sum over training examples in one task

Meta Learning

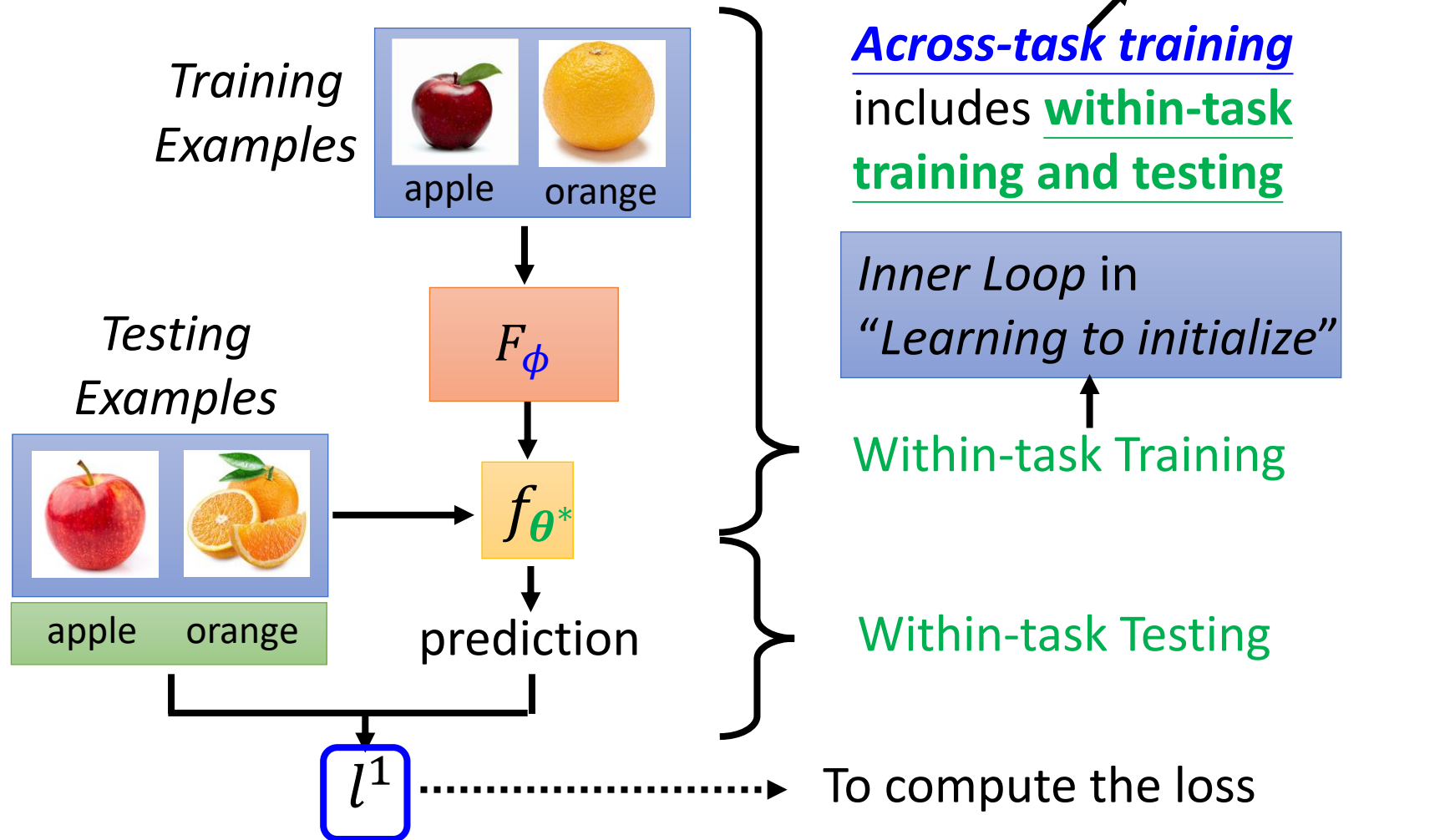
$$L(\phi) = \sum_{n=1}^N l^n$$

Sum over testing examples in one task

Sum over training tasks

$$L(\phi) = \sum_{n=1}^N l^n$$

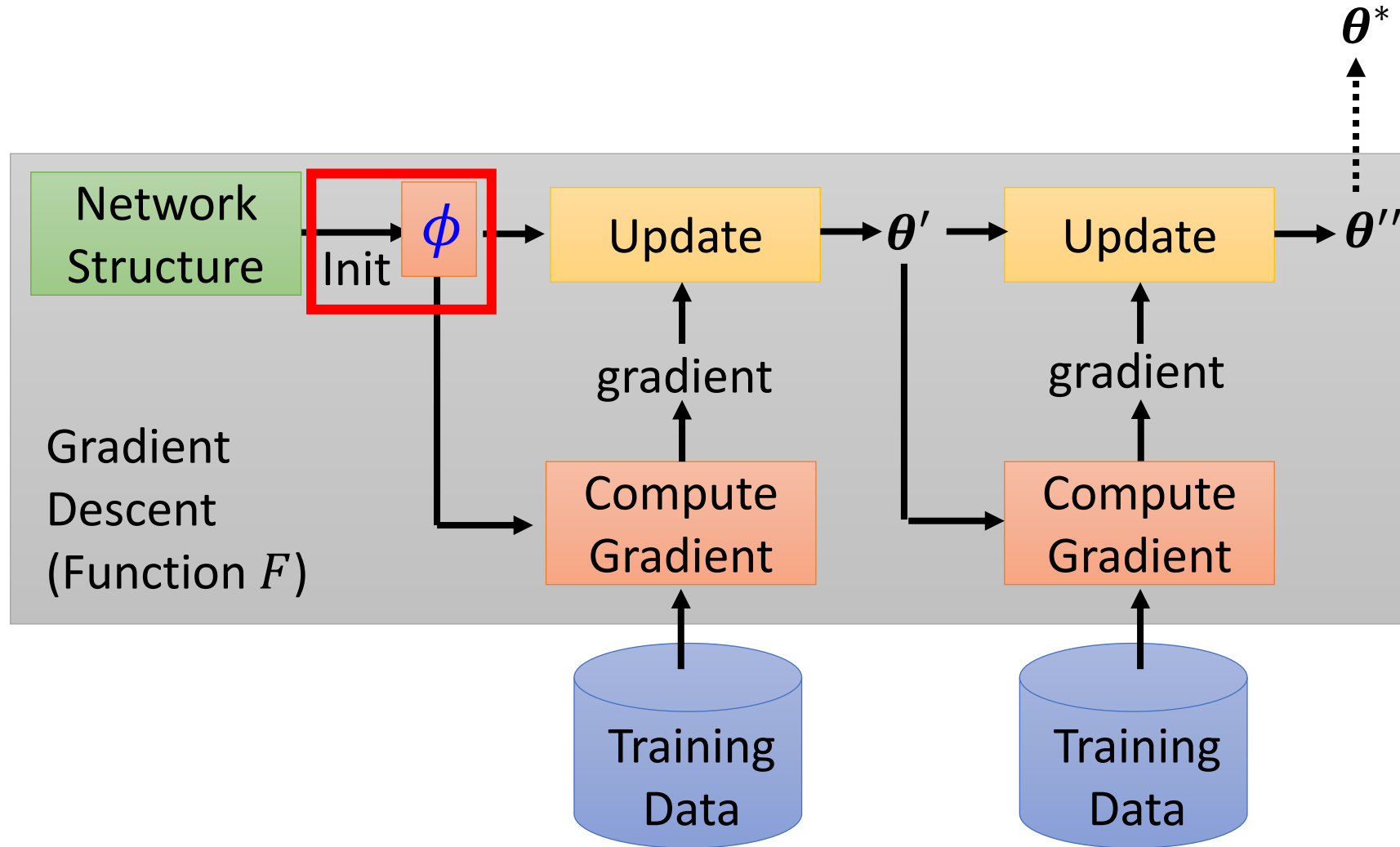
If your optimization method needs to compute $L(\phi)$



Meta Learning v.s ML

- What you know about ML can usually apply to meta learning
 - Overfitting on training tasks
 - Get more training tasks to improve performance
 - Task augmentation
 - There are also hyperparameters when learning a learning algorithm
 - Development task 😊

Review: Gradient Descent



Learning to initialize

- Model-Agnostic Meta-Learning (MAML)



Mammals

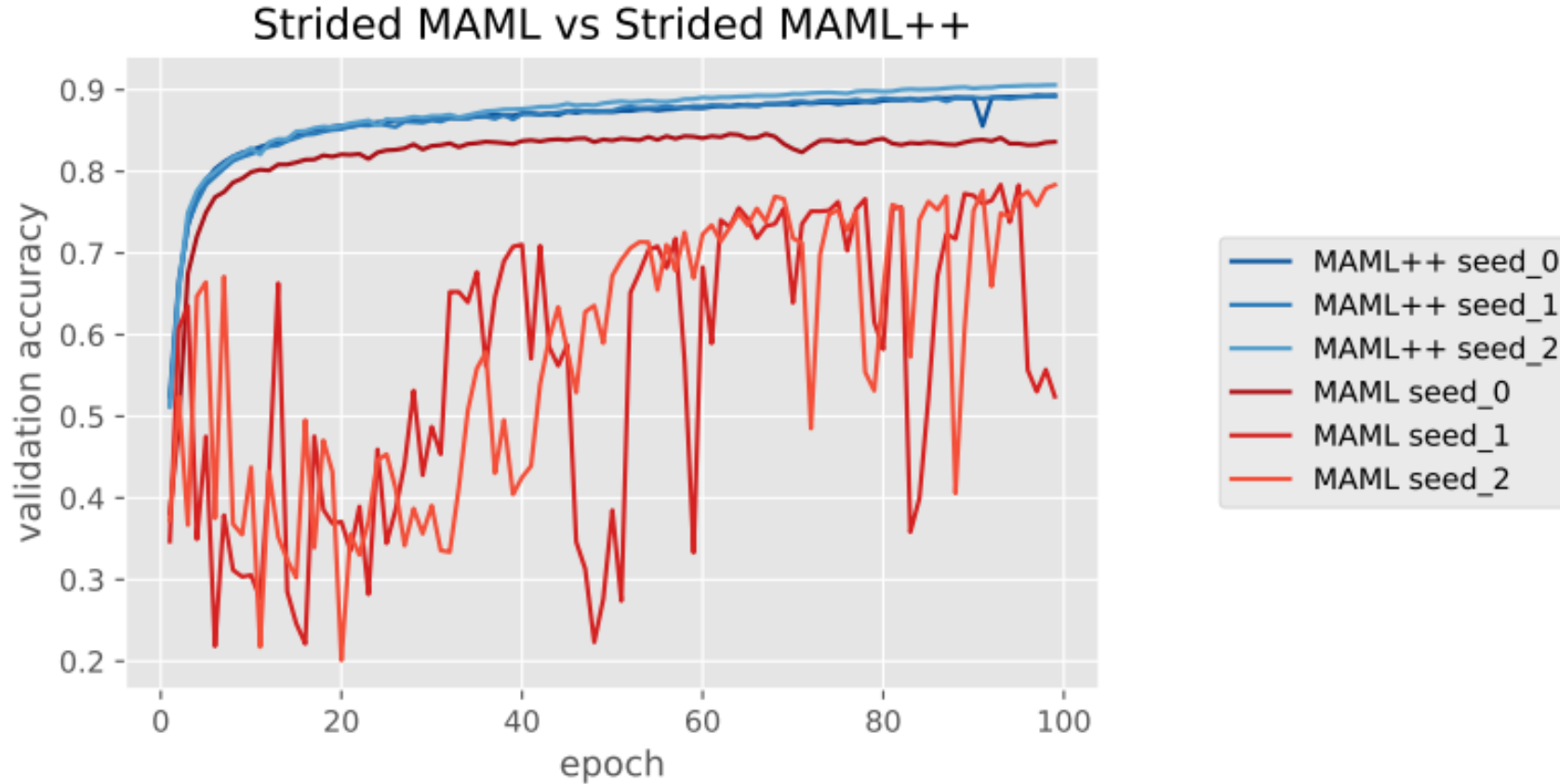
Chelsea Finn, Pieter Abbeel, and Sergey Levine, “Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks”, ICML, 2017

- Reptile



<https://arxiv.org/abs/1803.02999>

How to train your ~~Dragon~~ MAML



Antreas Antoniou, Harrison Edwards, Amos Storkey, How to train your MAML, ICLR, 2019

MAML



Pre-training (Self-supervised Learning)



MAML

Isn't it domain adaptation / transfer learning?



Pre-training (more typical ways)



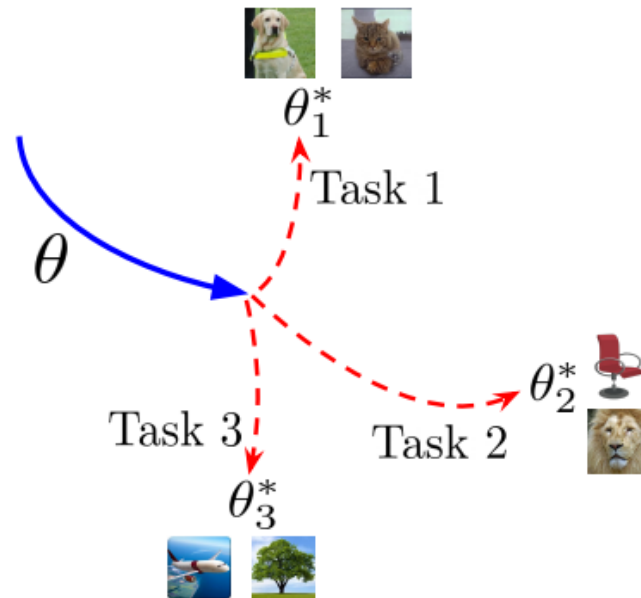
Use data from different tasks
to train a model

Also known as multi-task
learning (baseline of meta)

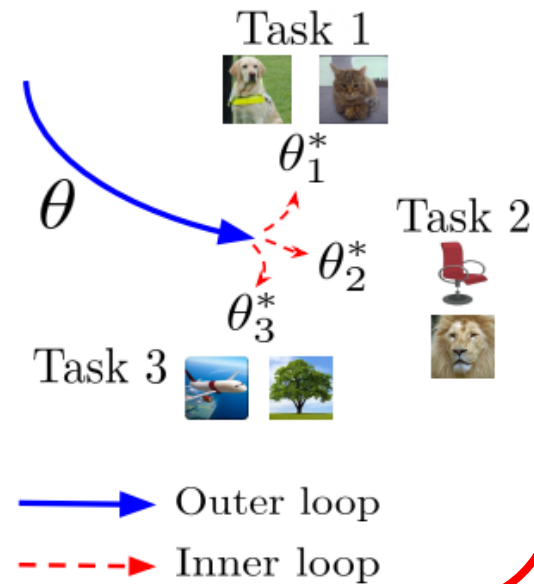
MAML is good because

- ANIL (Almost No Inner Loop)

Rapid Learning



Feature Reuse



Aniruddh Raghu, Maithra Raghu, Samy Bengio, Oriol Vinyals, Rapid Learning or Feature Reuse? Towards Understanding the Effectiveness of MAML, ICLR, 2020